

BUSA8001 – APPLIED PREDICTIVE ANALYTICS

PROGRAMMING TASK 2

APPLYING KMEANS++ AND AGGLOMERATIVE CLUSTERING TECHNIQUES FOR CUSTOMER SEGMENTATION WITH PROPOSED MARKETING STRATEGIES

ID: 48007900

October 2024

BUSA8001 – Applied Predictive Analytics

Contents

I.	Introduction	3
II.	Exploratory Data Analysis (EDA)	3
1.	Data Overview	3
2.	Univariate Analysis - Distributions of each variable	4
3.	Bivariate Analysis – Correlation Matrix between variables	6
III.	Customer Segmentation	7
1.	Standardisation of Variables.....	7
2.	Determining Optimal Number of Clusters	7
3.	Clustering Techniques	8
4.	Profiling customer segmentation	8
IV.	Recommendations: Tailored Marketing Strategies for Identified Customer Segments.....	9
	Cluster 1: Middle-aged, Wealthy, Predominantly Female, Married, Living in Large Cities.....	9
	Cluster 2: Younger, Lower-Income, Predominantly Male, Single, Living in Smaller Cities	10
V.	Conclusion.....	10

I. Introduction

In travel industry, recognising customer needs is essential for the development effective marketing strategies. This report aims to address the challenge of identifying distinct customer segments through the implementation of clustering techniques. To achieve this objective, an Exploratory Data Analysis (EDA) was conducted to explore and understand the distributions and relationships of the data. Then, two clustering techniques, KMeans++ and Agglomerative were utilised to divide customers into distinct segments. Lastly, the identified segments were interpreted through tables with provided marketing strategies to maximise customer engagement.

II. Exploratory Data Analysis (EDA)

1. Data Overview

The dataset consists of 2000 records and 7 variables. All columns are in integer datatype and do not contain any missing values (see Figure 1).

```
#      Column      Non-Null Count  Dtype
---  -
0     Gender      2000 non-null    int64
1     Marital Status  2000 non-null    int64
2     Age          2000 non-null    int64
3     Education     2000 non-null    int64
4     Income        2000 non-null    int64
5     Occupation    2000 non-null    int64
6     Settlement Size 2000 non-null    int64
dtypes: int64(7)
memory usage: 109.5 KB
```

Figure 1. Basic Information of the Dataset

The Table 1 below classifies each feature based on its variable type.

Variable type	Features list	Explanation
Numeric	Gender, Marital Status, Occupation, Settlement Size	Categorical, no inherent order
Ordinal	Education	Categories with an order
Nominal	Age, Income	Numeric (continuous)

Table 1. Classification of Encoded Variables by Type

2. Univariate Analysis - Distributions of each variable

Univariate analysis aims to examine the distribution of a single variable. This provides insights into the behaviour of individual variables before exploring relationship between multiple variables.

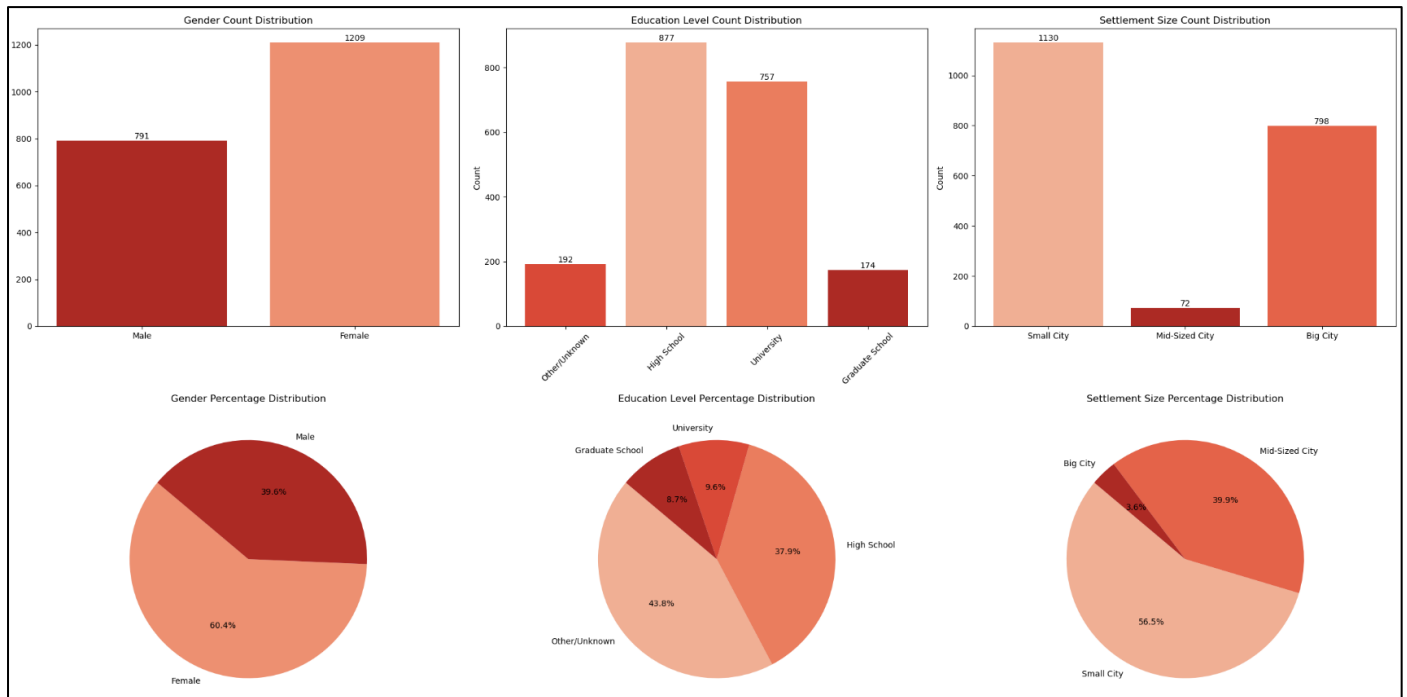


Figure 2. Visualisation of Categorical Data: Gender, Education Level, and Settlement Size

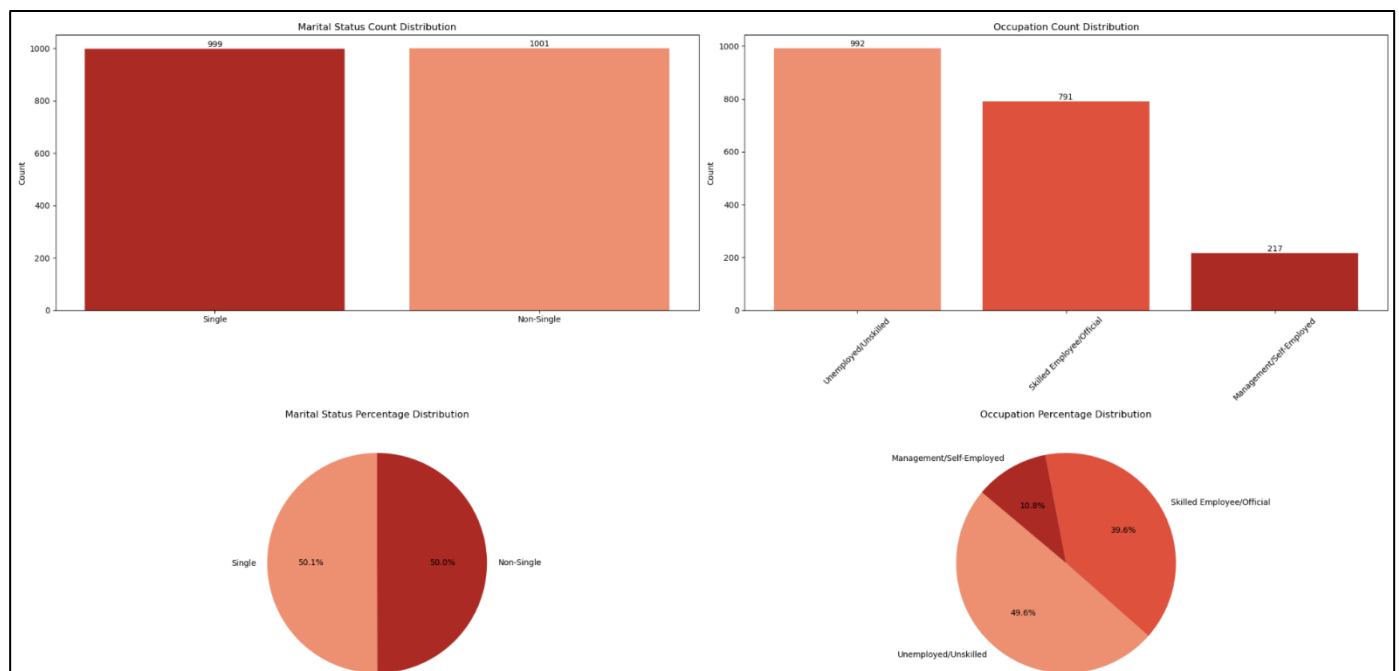


Figure 3. Visualisation of Categorical Data: Marital Status and Occupation

a. Summary of Categorical Data Distributions (Figure 2 & 3)

1. **Gender:** 60.4% are female, and 39.6% are male.
2. **Education:** The majority is in high school (43.8%) or university education (37.9%). A smaller proportion has graduate degrees (8.7%) or "Other/Unknown" (9.6%).
3. **Settlement Size:** Most individuals live in small cities (56.5%), followed by big cities (39.9%), with only 3.6% from mid-sized cities.
4. **Marital Status:** Almost equally split between single (50.1%) and non-single (50.0%).
5. **Occupation:** Nearly half of individuals are unemployed or unskilled (49.6%), with 39.6% skilled workers and 10.8% in management or self-employed roles.

b. Summary of Numeric Data Distributions (Figure 4)

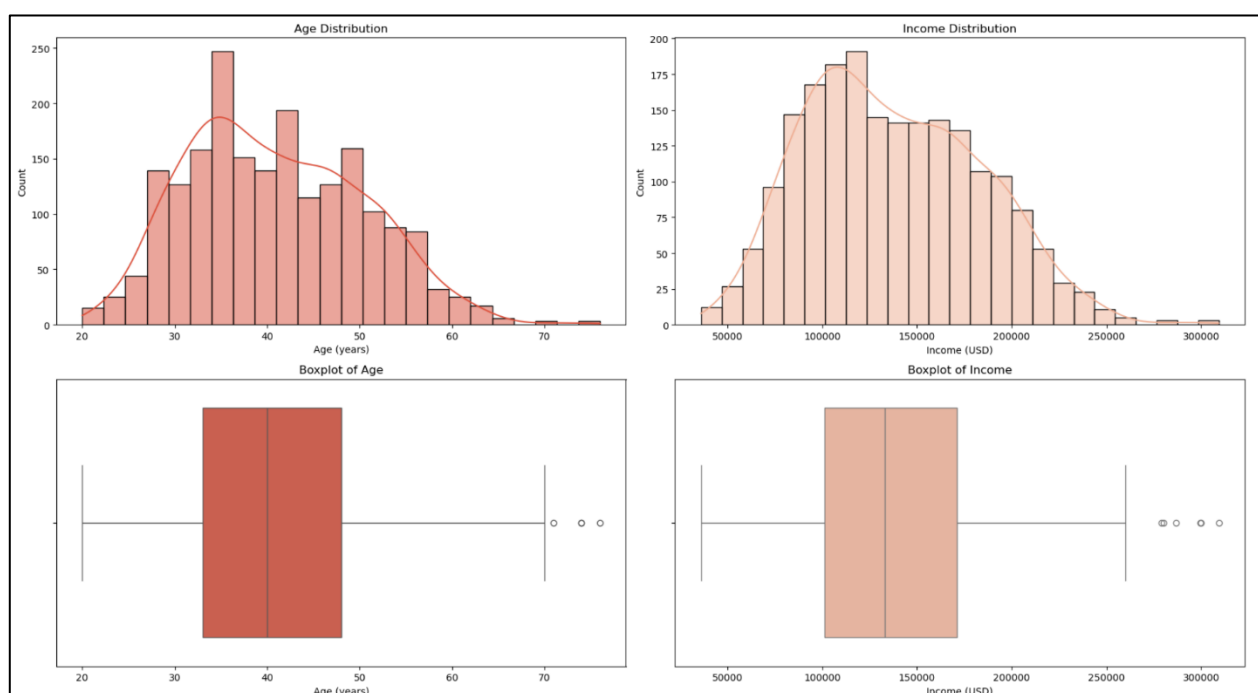


Figure 2. Visualisation of Numeric Data: Age and Income

1. **Age:** The age distribution is slightly normal, with most individuals between 30 and 50 years old. The boxplot shows a median age of about 40 years.
2. **Income:** The income distribution is slightly right skewed, with most individuals earning between \$50,000 and \$150,000. The boxplot indicates a median income around \$120,000

There are several of outliers in the boxplot for both age and income distribution, though not significant. These outliers can be remained in the dataset as it may represent real world case, such as wealthy individuals who are above 70.

3. Bivariate Analysis – Correlation Matrix between variables

The Correlation Matrix below visualises the strength and direction of relationships between two variables, providing insights into potential dependencies within the dataset.

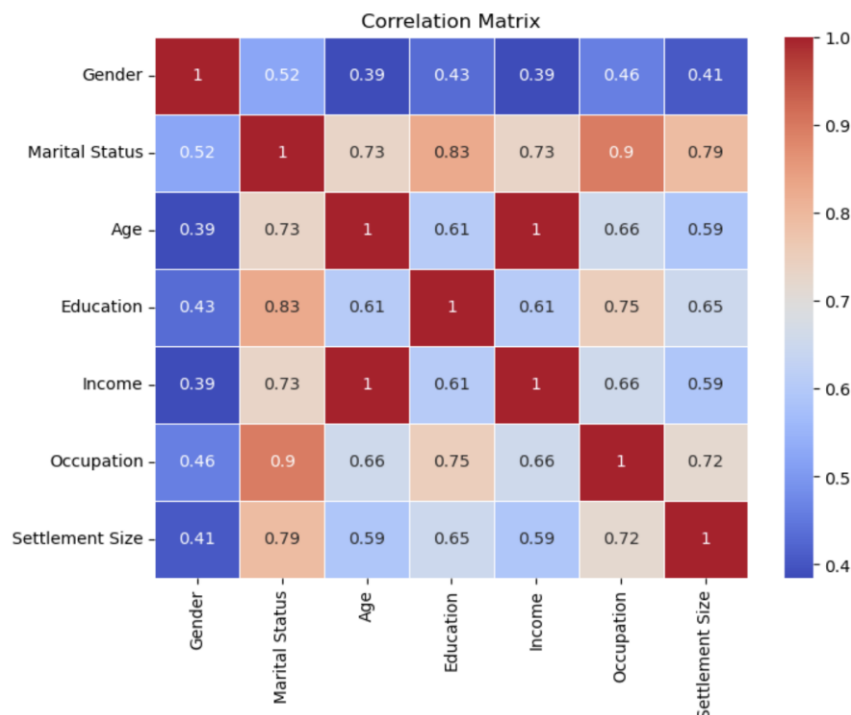


Figure 3. Correlation Matrix of the dataset

Key Insights:

1. **Age and Income** ($r=1.00$): Perfect positive correlation, indicating a strong linear relationship where income increases with age.
2. **Marital Status and Occupation** ($r=0.90$): Strong positive correlation, indicating that non-single individuals tend to hold better career prospects.
3. **Marital Status and Education** ($r=0.83$): Strong positive correlation, indicating that higher education levels are more likely to be non-single.
4. **Education and Occupation** ($r=0.75$): High positive correlation, indicating that higher education levels are associated with more advanced occupations.
5. **Settlement Size and Occupation** ($r=0.72$): Moderate positive correlation, indicating that individuals in larger cities are more likely to hold skilled jobs.

Overall, the dataset represents a set of individuals, mainly aged between 30 and 50, with moderate education and income levels, a significant proportion are unemployed or in unskilled jobs. There are clear relationships between socio-economic factors such as education, occupation and marital status, suggesting interdependency in determining individual social's status.

III. Customer Segmentation

1. Standardisation of Variables

To ensure no variable has a disproportionate influence when applying clustering methods, numeric variables (Age and Income) were standardised. This step was crucial as these numeric variables have different units and scales.

```
from sklearn.preprocessing import StandardScaler

df_scaled = df.copy()

# Standardising the numeric columns using StandardScaler
scaler = StandardScaler()
df_scaled[['Age', 'Income']] = scaler.fit_transform(df[['Age', 'Income']])
```

2. Determining Optimal Number of Clusters

a. The Elbow method:

The Elbow Method was used to determine the optimal number of clusters. Initially, k=2 was chosen as the optimal point where the decreasing rate of Cluster Inertia starts to flatten, suggesting that two clusters provide the best balance between variance reduction and simplicity.

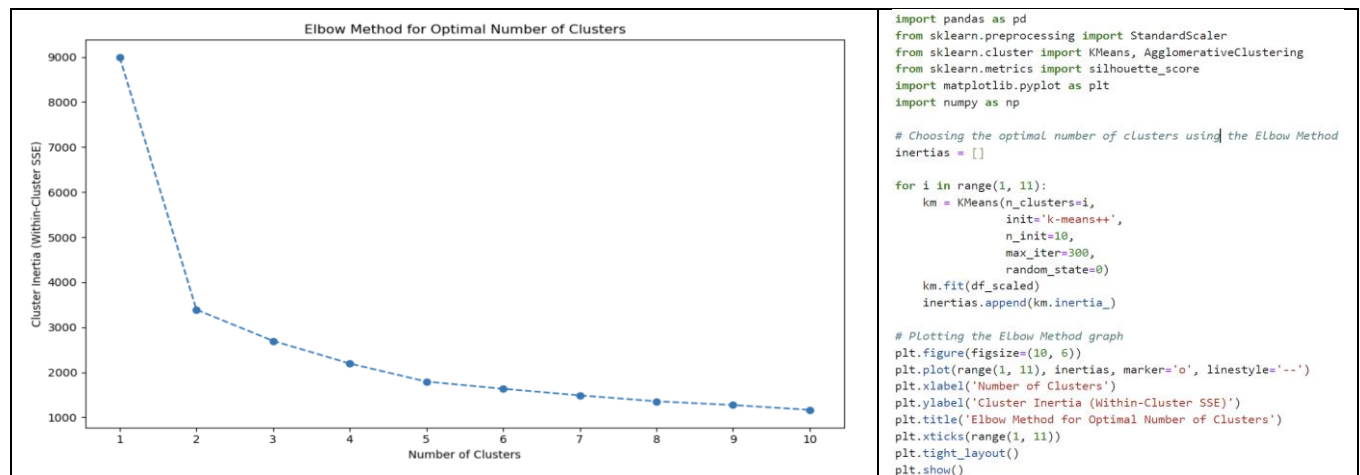


Figure 4. Visualisation of Elbow graph and inputs

b. Silhouette Plots Analysis for k=2:

To validate the choice of k=2, **Silhouette Plots Analysis** was performed:

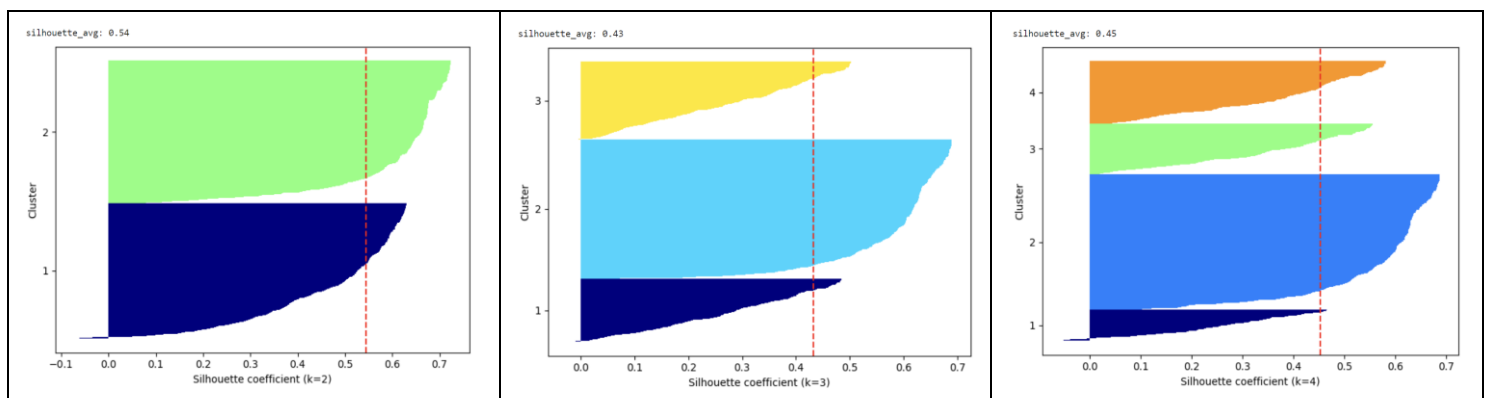


Table 2. Silhouette Averages and Plots for k=2, k=3, and k=4, respectively

Among the three plots, k=2 has the most defined bands, each extending towards the right (closer to 1). Also, the consistent width across each band reflects that most data points are better assigned to their respective clusters, leading to the high average silhouette score of 0.54. Overall, k=2 was selected for its consistent shape and extension toward the right, leading to high average score.

3. Clustering Techniques

The application of k=2 in K-Means++ and Agglomerative clustering was estimated with all variables to segment the customers into two distinct groups. Figure 5 and Figure 6 illustrate the cluster centers for all variables as well as the number of customers in each cluster.

i. K-means++ Clustering

	Gender	Marital Status	Age	Education	Income	Occupation	Settlement Size	Customer Count
0	0.856410	0.992821	48.174359	2.103590	173443.318974	1.215385	1.636923	976
1	0.364878	0.032195	33.831220	0.840976	103341.616585	0.039024	0.070244	1024

Figure 5. KMeans++ Clustering Summary (k=2)

ii. Agglomerative Clustering

	Gender	Marital Status	Age	Education	Income	Occupation	Settlement Size	Count
0	0.774721	0.856406	47.080825	1.925193	168085.081685	1.043852	1.384351	1163
1	0.367981	0.005974	32.129032	0.805257	95041.150538	0.013142	0.069295	837

Figure 6. Agglomerative Clustering Summary (k=2)

iii. Interpretation of each Clustering Techniques

Cluster	KMeans++ Clustering (k=2):	Agglomerative Clustering (k=2)
Cluster 1	• Middle-aged individuals with an average age around 48.2 years old. • Having high average income about \$173,443. • Most members are female (86%) and being married (99%). • High tendency of residing in large cities and having high level of education and occupations.	• Middle-aged individuals with an average age around 47.1 years old. • Having high average income about \$168,085. • Most members are female (77%) and being married (86%). • High tendency of residing in large cities and having high level of education and occupations.
Cluster 2	• Young individuals with an average age around 33.8 years old. • Having relatively low income about \$103,341. • Most members are male (64%) and being single (97%). • High tendency of residing in small cities and having low level of education and occupations.	• Young individuals with an average age around 32.1 years old. • Having relatively low income about \$95,041. • Most members are male (63%) and being single (99%). • High tendency of residing in small cities and having low level of education and occupations.

Table 3. Interpretation of KMeans++ and Agglomerative Clusters

4. Profiling Customer Segmentation

Based on the cluster interpretation, both K-Means++ and Agglomerative techniques provide two well-defined clusters:

Cluster 1: Middle-aged individuals residing in larger cities with high-income, being married and having high level of education and occupations.

Cluster 2: Younger individuals residing in small cities with low-income, being single and having low level of education and occupations.

The clusters from both techniques are generally overlapping with minor differences in the outputs, indicating that both techniques provide consistent customer segments. This allows the development of innovative marketing strategies that effectively maximise the engagement of each cluster.

IV. Recommendations: Tailored Marketing Strategies for Identified Customer Segments

Cluster 1: Middle-aged, Wealthy, Predominantly Female, Married, Living in Large Cities

a. Exclusive Membership and Loyalty Programs:

- i. **Strategy:** Attract and retain high-income customers by offering VIP membership programs with personalised services.
- ii. **Approach:** Offer personalised deals and loyalty points system that can be redeemed for exclusive upgrades and services.
- iii. **Example:** Launch “Luxury Travel Club” with benefits such as early booking access, additional services and points exchange.

b. Family Travel Packages:

- i. **Strategy:** Create travel packages that appeal to family travel needs.
- ii. **Approach:** Create packages designed for families with activities suitable for all age groups.
- iii. **Example:** Promote family vacation bundles that offer deals on family vacations, offering both kid-friendly activities and adult experiences.

Cluster 2: Younger, Lower-Income, Predominantly Male, Single, Living in Smaller Cities

a. Group and Solo Travel Discounts:

- i. **Strategy:** Encourage group travel and offer discount for solo travellers.
- ii. **Approach:** Implement referral programs where customers can earn discounts for bringing friends or booking solo adventures.
- iii. **Example:** Launch promotion campaigns that encourage customers to bring friends or book solo adventures to earn travel perks.

b. Budget-Friendly Adventures:

- i. **Strategy:** Offer low-cost travel options with a focus on adventures and exploration.
- ii. **Approach:** Create packages that bundle affordable living, transport and activities.
- iii. **Example:** Collaborate with low-cost airlines and economy hotels to offer discounts upon booking.

V. Conclusion

This report aimed to segment customers by applying KMeans++ and Agglomerative clustering techniques to develop effective marketing strategies for the travel agency. The dataset includes demographic and socio-demographic attributes such as Age, Income, and Education level. The methodology involved scaling numeric features and applying both clustering techniques to identify distinct segments. The analysis revealed two distinct segments: Cluster 1 consists of middle-aged, wealthy, married individuals in large cities, while Cluster 2 consists of younger, lower-income, single individuals in smaller cities. Tailored marketing strategies were proposed, including membership and family-focused packages for Cluster 1 and budget-friendly adventures for Cluster 2. This segmentation provides insights into the customer base, aiding to develop better targeted marketing campaigns and maximise customer engagement and sales growth.

Word count: 1062 words (excluding headings, tables and figures)